



Rounding off

THE PROCESS OF ROUNDING OFF INVOLVES REPLACING ONE NUMBER WITH ANOTHER TO MAKE IT MORE PRACTICAL TO USE.

SEE ALSO

◀ 44–45 Decimals

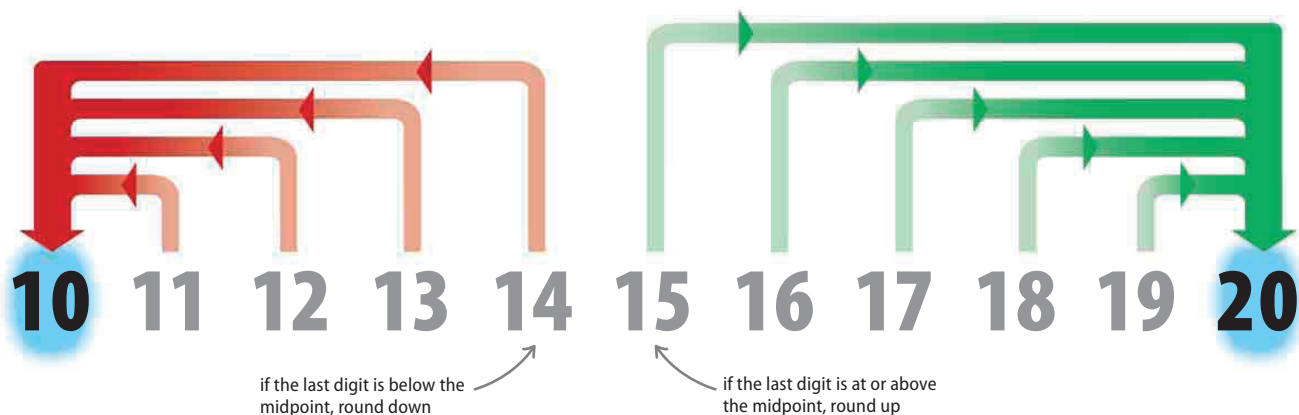
◀ 66–67 Mental math

Estimation and approximation

In many practical situations, an exact answer is not needed, and it is easier to find an estimate based on rounding off (approximation). The general principle of rounding off is that a number at or above the midpoint of a group of numbers, such as the numbers 15–19 in the group 10–20, rounds up, while a number below the midpoint rounds down.

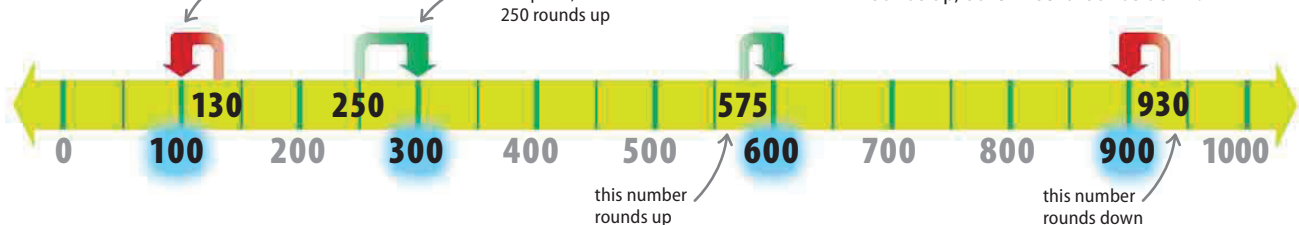
▽ Rounding to the nearest 10

The midpoint between any two 10s is 5. If the last digit of each number is 5 or over it rounds up, otherwise it rounds down.



▽ Rounding to the nearest 100

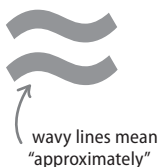
The midpoint between two 100s is 50. If the second digit is 5 or over, the number rounds up, otherwise it rounds down.



LOOKING CLOSER

Approximately equal

Many measurements are given as approximations, and numbers are sometimes rounded to make them easier to use. An “approximately equals” sign is used to show when numbers have been rounded up or down. It looks similar to a normal equals sign (=) but with curved instead of straight lines.



$$31 \approx 30 \quad \text{and} \quad 187 \approx 200$$

△ Approximately equal to

The “approximately equals” sign shows that the two sides of the sign are approximately equal instead of equal. So 31 is approximately equal to 30, and 187 is approximately equal to 200.